

PROJECT REPORT

SMART STUDY TIME RECOMMENDER USING AI

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PROBLEM STATEMENT

Students often find it difficult to decide how much time to dedicate to different subjects while preparing for exams. In many cases, study time is distributed randomly or based on intuition rather than any structured approach. As a result, important or difficult subjects may not receive enough attention, leading to inefficient preparation.

This project aims to address this issue by developing a system that recommends how many hours a student should study for each subject based on multiple real-world factors.

WHY THIS PROBLEM MATTERS

Time management is one of the most common challenges students face. When study time is not allocated effectively, it can lead to incomplete syllabus coverage, increased stress close to exams, and ultimately lower academic performance.

A system that helps students make informed decisions about how to distribute their study time can significantly improve both efficiency and outcomes. Since this problem is closely tied to everyday student life, solving it has clear practical value.

APPROACH TO SOLVING THE PROBLEM

To solve this problem, an AI-based approach was used.

The system takes input parameters such as:

- subject difficulty
- preparation level
- importance of the subject

- number of days left before the exam

Based on these inputs, the system predicts how many hours a student should ideally spend studying each subject.

A Decision Tree Regressor was chosen as the machine learning model. It is simple, easy to interpret, and works well with smaller datasets. The model was trained on a manually created dataset designed to reflect realistic study scenarios.

However, relying only on the model was not sufficient. During testing, it became clear that some predictions needed refinement. To address this, a rule-based adjustment layer was introduced. For instance, study time is increased when the exam is very close and slightly reduced when preparation levels are already high.

The system also supports multiple subjects and generates a structured study plan by organizing and prioritizing them based on the required study time

KEY DECISIONS MADE

Several important decisions were made during the development of this project:

- Using a Decision Tree model: Chosen for simplicity and interpretability
- Creating a custom dataset: Since real data was not available, a small dataset was designed manually
- Combining ML with rule-based logic: This helped to ensure predictions were more realistic
- Supporting multiple subjects: This made the system more practical and useful as students often have multiple subjects

- **Sorting output:** Helps users prioritize subjects effectively

These decisions helped balance simplicity, accuracy, and usability

IMPLEMENTATION OVERVIEW

The project was implemented using Python.

Libraries used:

- Pandas (for data handling)
- Scikit-learn (for machine learning)

The program follows this flow:

1. Take number of subjects as input
2. Collect details for each subject
3. Predict study hours using the trained model
4. Apply rule-based adjustments
5. Assign priority levels and advice
6. Display a structured and sorted study plan

CHALLENGES ENCOUNTERED

During the development of the project, several challenges were encountered:

- Designing a realistic dataset without real-world data
- Ensuring predictions were not too high or too low
- Balancing machine learning with rule-based logic
- Handling multiple subject inputs efficiently
- Making the output clear and structured

To address these challenges, multiple rounds of testing and refinement were carried out.

Initially, the model was trained on a small manually created dataset. However, early testing revealed that some predictions were unrealistic, especially in extreme cases such as very high difficulty with very low preparation. To improve this, the dataset was adjusted by adding more balanced and varied data points to better represent real study scenarios.

The model was then tested with different combinations of inputs, including edge cases such as:

- very few days left before the exam
- very high preparation levels
- low-importance subjects

These tests helped identify inconsistencies in predictions. To correct this, a rule-based adjustment layer was introduced. For example, study time was increased when the exam was very close and reduced when preparation was already high. This significantly improved the practical accuracy of the results.

Another challenge was handling multiple subjects. Initial versions worked only for a single subject, so the program was redesigned to loop through multiple inputs and store results in a structured format. Additional testing ensured that the program correctly handled multiple entries without errors.

Finally, improvements were made to the output format by organizing results into a sorted table, prioritizing subjects based on required study time. This made the output more readable and useful for real-world application.

Overall, iterative testing and gradual improvements helped refine both the model and the program, resulting in a more reliable and user-friendly system.

WHAT I LEARNED

Through the course of this project, I developed a clearer understanding of how theoretical AI concepts translate into practical applications.

Working with a Decision Tree model helped me understand how machines can learn patterns from data and make predictions based on input features. At the same time, I realized that relying solely on machine learning is not always sufficient. Incorporating rule-based logic made the system more grounded and aligned with real-world expectations.

I also gained insight into the importance of data quality. Since the dataset was manually created, even small changes in values had a noticeable impact on predictions. This highlighted how crucial balanced and representative data is in any AI system.

Beyond technical concepts, the project improved my ability to think through problems step by step, test different scenarios, and refine a solution iteratively. It reinforced the idea that building an effective system is rarely a one-step process—it evolves through continuous testing, observation, and improvement.

CONCLUSION

The Smart Study Time Recommender demonstrates how artificial intelligence can be applied to solve a simple yet meaningful real-world problem. By combining machine learning with logical

adjustments, the system is able to generate structured and practical study recommendations.

What makes this project effective is not just the use of AI, but the way it integrates different approaches to arrive at realistic outputs. It shows that even with a small dataset and basic models, useful solutions can be developed when the problem is clearly understood.

Overall, the project highlights how AI can move beyond theory and be applied in everyday contexts, making it both functional and relevant.

FUTURE IMPROVEMENTS

The project can be further improved by:

- using real student datasets
- developing a graphical user interface (GUI)
- adding daily timetable generation
- deploying it as a web or mobile application